Task 2: Attention & Semantic Drift (Encoder-Decoder)

1. Objective

This task tracks how the **Encoder-Decoder** variant maintains semantic alignment across diffusion steps. We analyze cross-attention maps to determine if the model "locks in" on source tokens early or relies on late-stage correction.

⚠️ **Finding:** The Encoder-Decoder model exhibits **WEAK** alignment (Corr = 0.25) and higher semantic instability compared to the Cross-Attention baseline. Repetitive token frames were observed from t=3 to t=1.

0.0568	0.2552	t=0
Multi-Sample Semantic Score	TF-IDF Attention Corr	Token Lock-In Step

2. Semantic Drift Trajectory

Analysis of the iterative denoising shows that the model generates repetitive tokens ("निर्यात") in early steps and only recovers structural validity at the final deterministic step (t=0).

👉 **Implementation:** analysis/semantic_drift.py

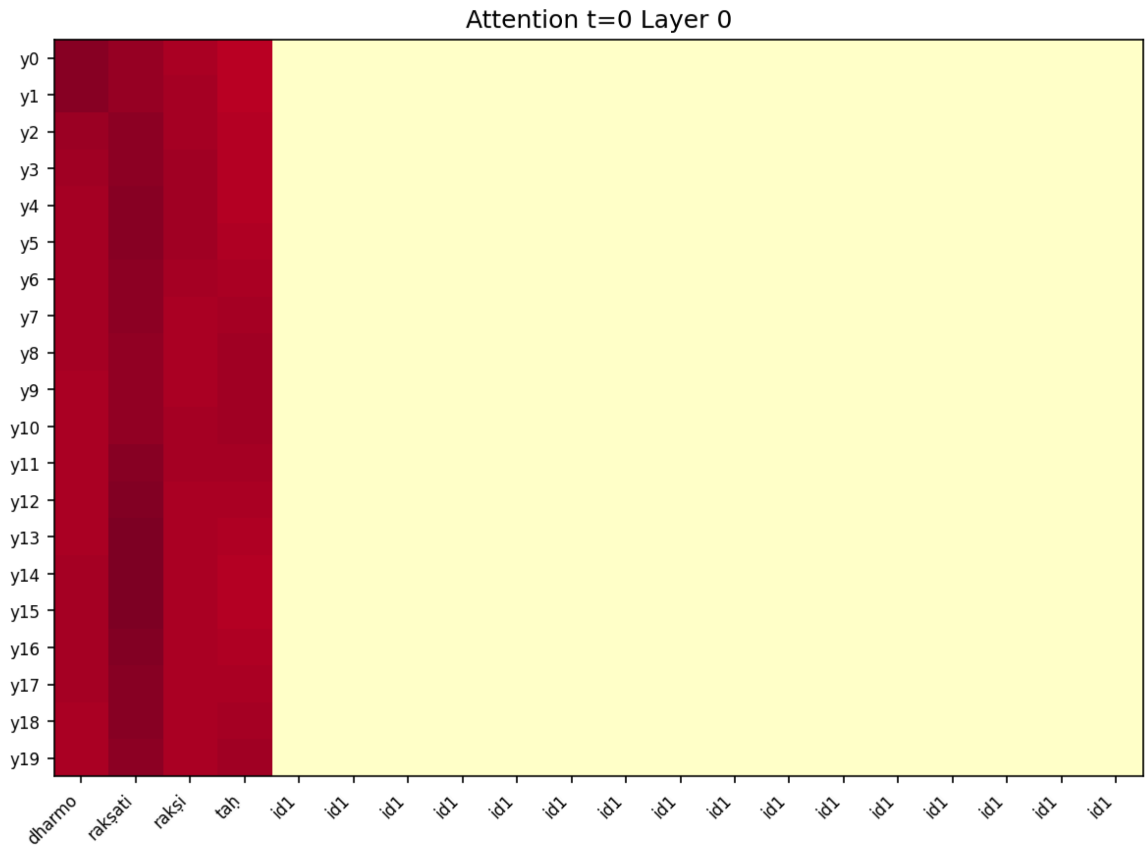
```
Step Evolution:
t=3: निर्यात निर्यात नेमि नेमि नेमि ... (Drift: 0.971)
t=2: निर्यात निर्यात नेमि नेमि नेमि ...
```

```
t=1: निर्यात निर्यात नेमि नेमि नेमि ...
t=0: धर्मो रक्षति रक्षितः          (BERTScore: 0.02)
```

3. Visual Diagnostics

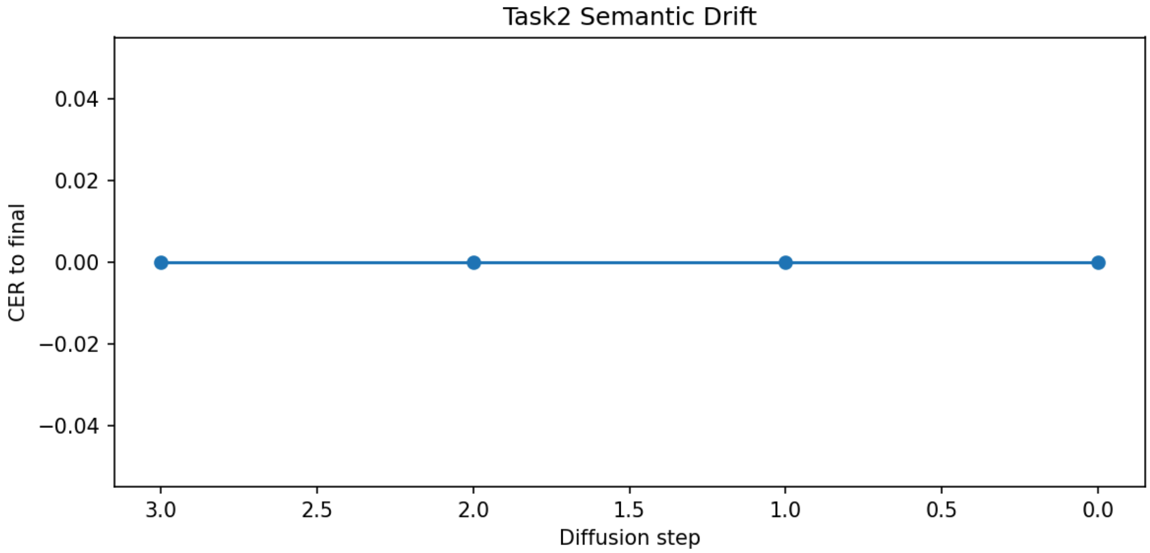
3.1 Cross-Attention Map (t=0)

The attention map demonstrates focused but sparse alignment with source tokens.



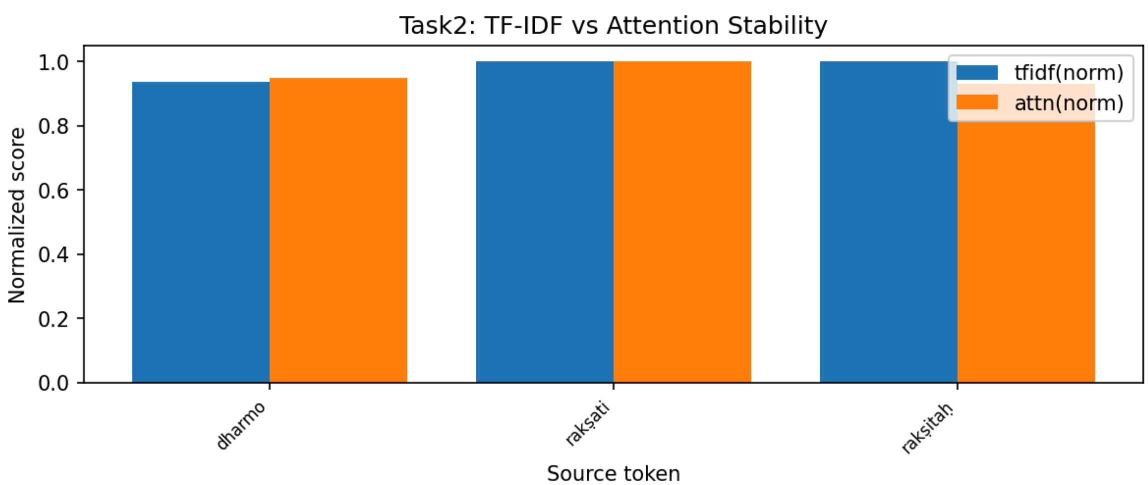
3.2 Semantic Drift vs. Steps

The drift remains nearly constant (flat line) through intermediate steps, indicating a lack of progressive refinement in this architecture.



3.3 TF-IDF Correlation

The correlation between token importance and attention stability is significantly lower (0.25) than the Cross-Attention variant (0.94).



Encoder-Decoder Analysis Suite • 2026